Mapping Mangrove Degradation and Coastal Connectivity

An exploratory SAR–optical and spatial machine-learning framework in Palawan, Philippines

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Abstract

Mangrove degradation often precedes complete canopy loss. This project explores a three-part framework combining radar and optical vegetation indicators, coastal proximity zones, and supervised land-cover classification. The implemented machine-learning benchmark is evaluated on temporally separated tabular samples; degradation and connectivity components remain demonstrations rather than validated ecological outcomes.

Scientific motivation

Optical greenness can remain high even when forest structure or tidal connectivity changes. Sentinel-1 backscatter and Sentinel-2 spectral indices could offer complementary information, while mangrove–reef proximity may help organize coastal monitoring priorities. Whether these indicators measure degradation and reef impacts requires field and hydrodynamic validation.

Research question

Can fused SAR–optical indices characterize degradation within standing mangroves, and how reliably can a coastal LULC classifier distinguish nearby land-use pressures?

Study area & spatial concept

Central Palawan, Philippines: 118.7°–119.0° E, 9.7°–10.0° N (Honda Bay / Puerto Princesa coastal setting). GIS buffering uses UTM Zone 51N (EPSG:32651).



Figure 1. Original portfolio GIS rendering of 2, 5 and 10 km buffer rings. Demonstration geometry, not observed ecological connectivity or modeled sediment transport.

Methods | research workflow

The three study components share a proposed five-feature SAR–optical representation, but their execution and evidence differ. The stages below reproduce the documented computational design.

Sentinel-1 SAR
VV + VH backscatter

Sentinel-2 MSI
Red + NIR + Green

1 · Feature derivation
RFDI = (VV − VH)/(VV + VH); NDVI = (NIR − Red)/(NIR + Red); NDWI = (Green − NIR)/(Green + NIR).

2 · A · Exploratory degradation score
Severity = RFDI × (1 − NDVI), min–max scaled within mangrove extent. Tested with synthetic raster arrays; not field-validated.

3 · B · Coastal proximity buffers
Degraded nuclei threshold > 0.5; 2 / 5 / 10 km buffers in EPSG:32651. Demo coordinates used because reef polygons were absent.

4 · C · Land-use classifier
Random Forest (300 trees), RBF-SVM and Gradient Boosting fitted on VV, VH, RFDI, NDVI and NDWI.

5 · Temporal holdout evaluation
Train: 16,001 samples tagged 2018–2022. Test: 3,000 samples tagged 2023–2024. Confusion matrix, accuracy, kappa and feature importance.

6 · Research outputs
CSV-based model evaluation, serialized RF model, GeoJSON buffer demonstration and a Leaflet/Chart.js research interface.

Methodological interpretation

The holdout measures classification of paddy, dryland agriculture and urban land cover only. A temporal label split alone does not establish independent field sampling or cross-region transfer. Geometric proximity buffers measure distance, not tidal or ecological connectivity.

Verified results | land-cover holdout

88.0%

Overall accuracy
2,640 / 3,000 correct · κ = 0.825

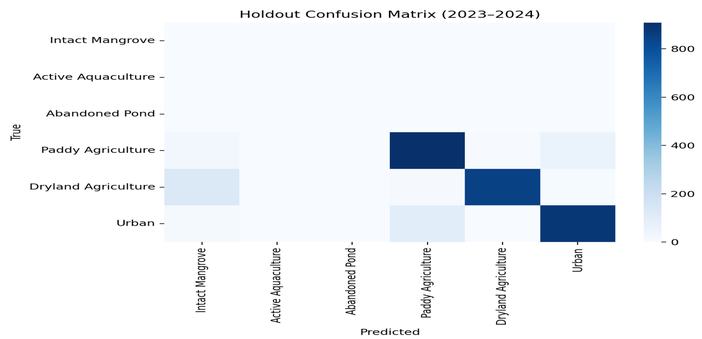


Figure 2. Actual project holdout confusion matrix. Mangrove and aquaculture classes are absent from the 2023–2024 reference set.

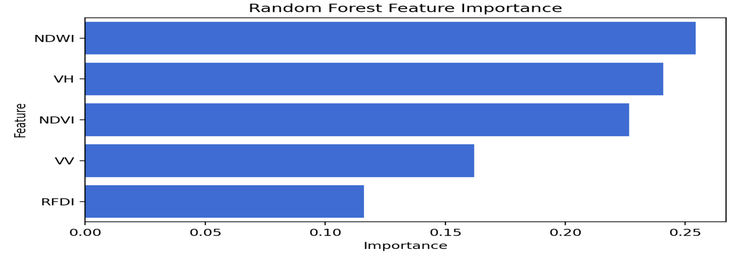


Figure 3. Verified Random Forest importance: NDWI, VH and NDVI are strongest predictors in this tabular evaluation; this is not proof of causal degradation drivers.

Interpretation and limitations

The classifier separates the three tested agricultural/urban categories, but the holdout cannot support performance claims for intact mangroves, active aquaculture or abandoned ponds. Degradation severity was computed from synthetic rasters. Reef buffers were produced with demonstration coordinates, without tidal-current or fish-biomass validation.

Next research steps

Acquire and calibrate real co-registered SAR–optical scenes; build a spatially independent reference dataset covering mangrove and pond classes; compare degradation indices with field canopy measurements; and replace circular buffers with hydrodynamic connectivity models.

Research status: implemented exploratory framework with a verified tabular holdout experiment. Coastal severity surfaces, reef connectivity and biomass effects remain unvalidated. Sources: supplied project audit, evidence matrix, and original repository figures (October 2026).